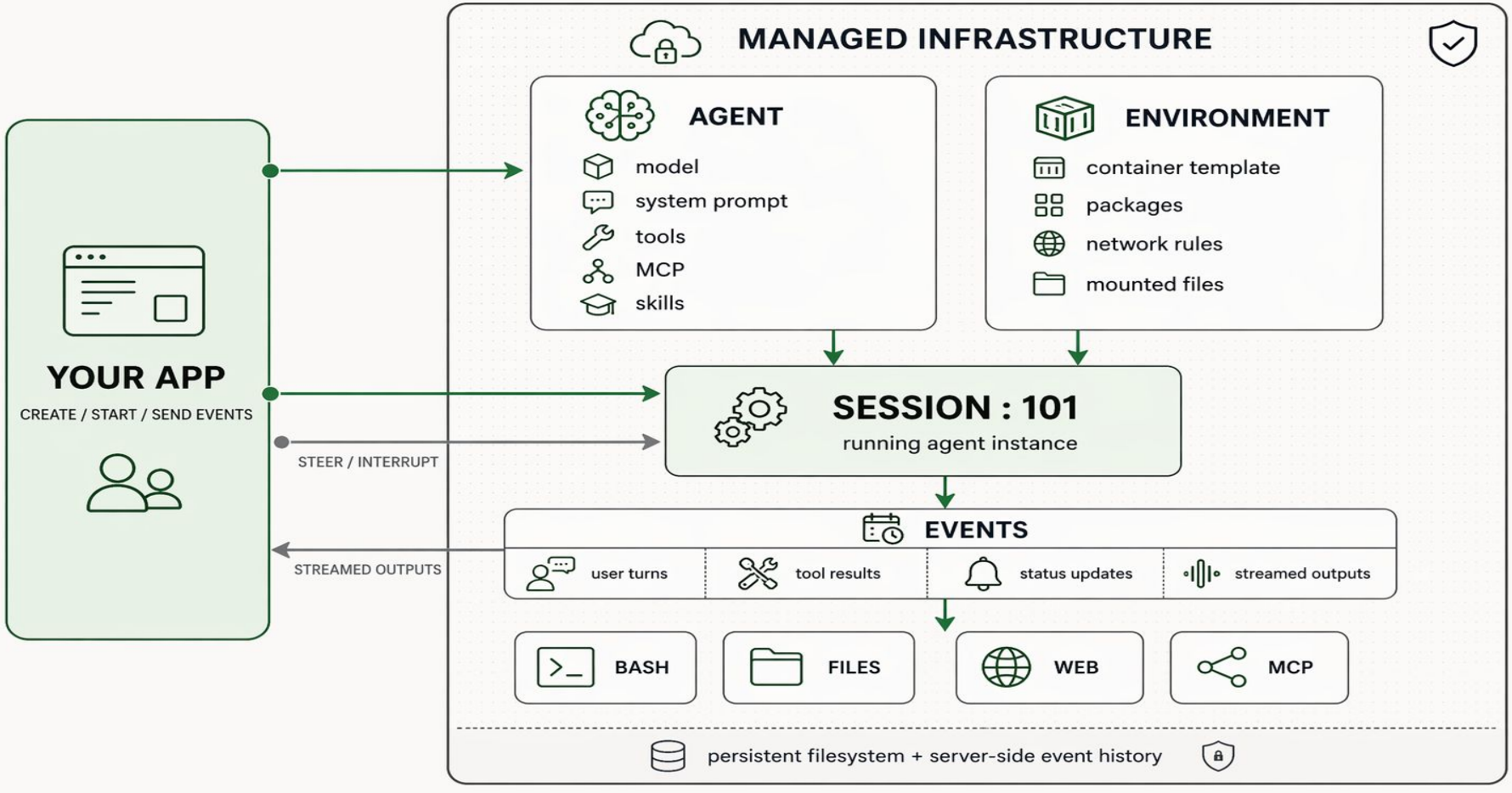
Managed Agents run long-running work in managed infrastructure



Define your agent as a reusable, versioned configuration

managed-agents-2026-04-01
beta header required

DEVELOPER / APP

- DEFINE ONCE
- REFERENCE LATER

AGENT

reusable versioned configuration

IDENTITY

- Agent ID (unique)
- Name
- Description
- Tags / Labels
- Owner / Team

CONFIGURATION (WHAT IT CAN DO)

MODEL
model provider, model name, parameters, context settings

SYSTEM PROMPT
persona, instructions, behavior guidelines, tone

TOOLS
built-in tools, custom tools, tool permissions, timeouts

MCP SERVERS
server list, capabilities, scopes, auth configuration

SKILLS
skill library, skill configs, execution strategy

METADATA
created by, created at, updated at, version, notes

POLICIES & GUARDRAILS

- Allowed Tools
- Network Rules
- Data Access
- Safety Limits
- Approval Rules
- Resource Limits

CAPABILITIES SUMMARY: Reason | Plan | Act | Use Tools | Learn

REFERENCE BY ID

SESSION

references agent by ID

- Starts execution
- Maintains state
- Streams events
- Can be steered or interrupted

VERSIONED

V1 Initial version created → **V2** Updated improvements → **V3** Further refinements → **ARCHIVED** Read only immutable

UPDATE → NEW VERSION

LIST VERSIONS | ARCHIVE = READ ONLY

WHY THIS MATTERS

REUSE
Define once, use everywhere

CONSISTENCY
Standardized behavior and capabilities

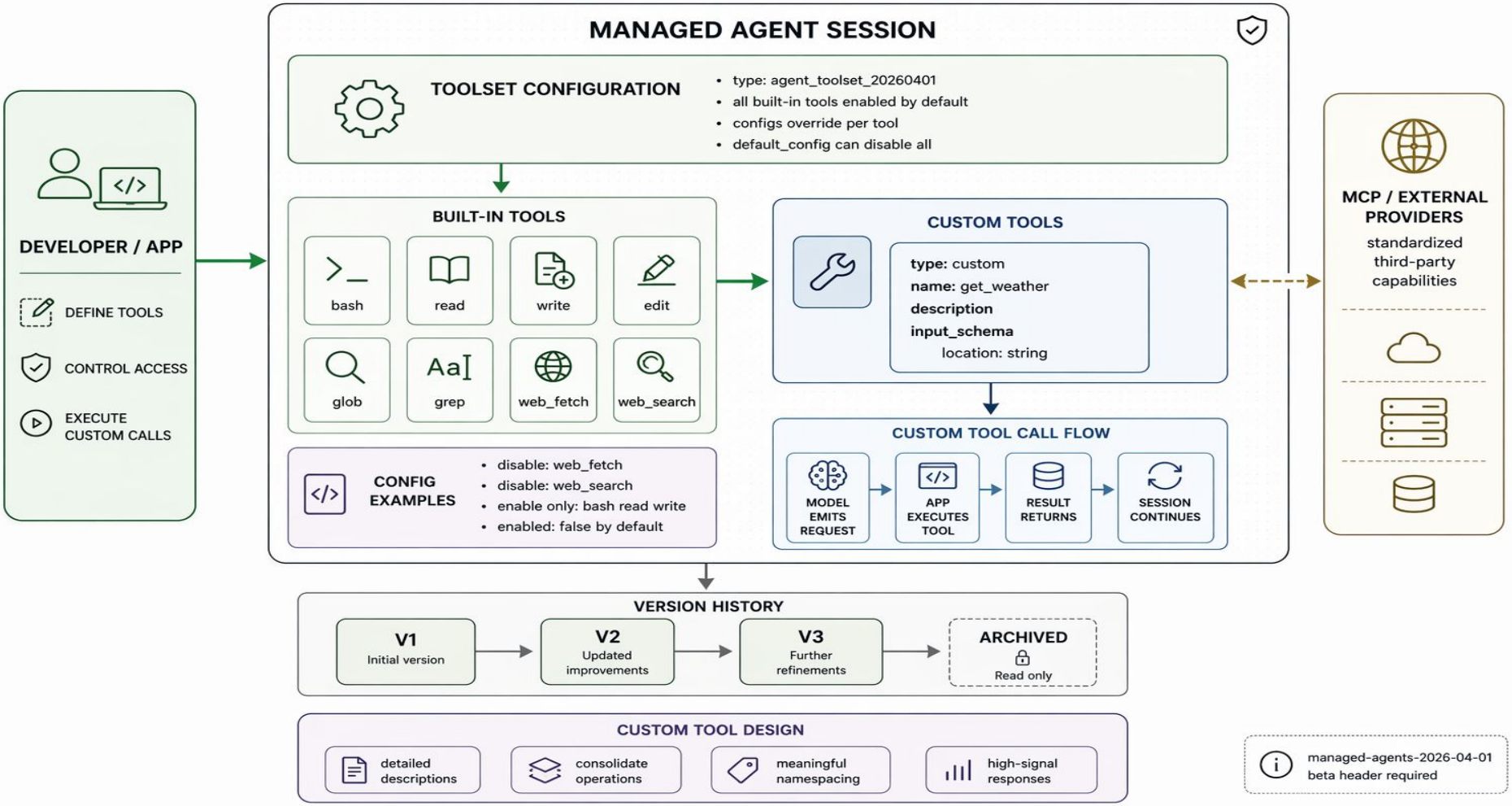
GOVERNANCE
Policies and guardrails enforce safety

EVOLUTION
Iterate and improve with versioning

LEGEND

- Create / Reference Flow
- Version / Lifecycle Flow
- Optional / Read-Only

Configure tools available to your agent



Configure MCP servers available to your agent

DEVELOPER / APP

CONFIGURE MCP SERVERS
Add, update, remove servers and scopes

CONTROL ACCESS
Restrict tools, resources and prompts per server

TRIGGER USAGE
Agent uses MCP servers during execution

MANAGED AGENT SESSION

MCP CONFIGURATION

- type: mcp_toolset_20260401
- define and attach MCP servers to the agent
- control which tools, resources, and prompts are exposed
- configs override per server
- default_config can disable all MCP servers

MCP SERVERS (CONNECTED TO AGENT)

SERVER	STATUS	TRANSPORT	SCOPES	ACTIONS
filesystem	Enabled	stdio	tools, resources	⋮
github	Enabled	sse	tools, resources, prompts	⋮
postgres	Enabled	stdio	tools	⋮
slack	Disabled	sse	tools, prompts	⋮

CONFIG EXAMPLES

- disable: slack
- scopes: [tools, resources] for filesystem
- allowed_tools: [read_file, list_dir] for filesystem
- default_config can disable all MCP servers

MCP INTERACTION MODEL

AGENT (MODEL)
decides when to use MCP based on task and context

MCP PROTOCOL
standardized messaging (JSON-RPC 2.0 over stdio / SSE)

MCP SERVER
exposes capabilities: tools, resources, prompts

EXECUTION CONTEXT
server executes and returns structured results

MCP TOOL CALL FLOW (FUNCTION CALL PATTERN)

1. **MODEL EMITS MCP REQUEST** — Chooses server, capability and args
2. **REQUEST SENT TO MCP SERVER** — Via protocol (stdio / SSE)
3. **SERVER EXECUTES THE CAPABILITY** — Per server logic and permissions
4. **RESULT RETURNS TO AGENT** — Structured response (JSON)
5. **SESSION CONTINUES** — Agent uses result in context

MCP / EXTERNAL PROVIDERS

standardized third-party capabilities

Files & Storage
Access and manage files and documents

Code & Dev Tools
Repositories, issues, CI/CD, etc.

Data & Databases
Query, read, write structured data

Communication
Message, notify, collaborate

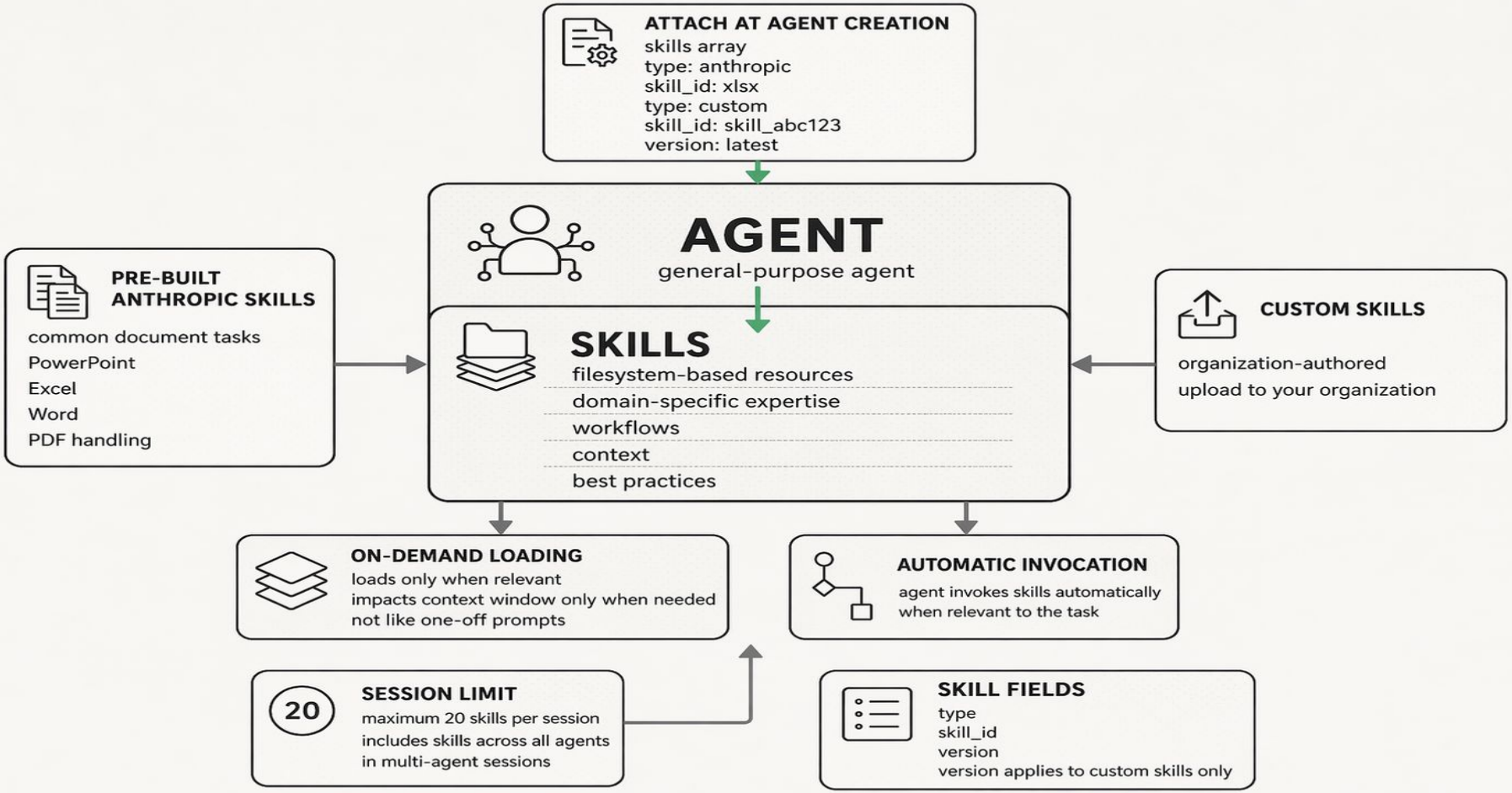
APIs & Services
Weather, maps, payments, etc.

MCP CONFIGURATION BEST PRACTICES

- Principle of least privilege
- Restrict scopes per server
- Allow only required tools/resources
- Review and audit server access
- Monitor usage and errors
- Version and update configurations

managed-agents-2026-04-01
beta header required

Attach reusable, filesystem-based expertise to your agent



managed-agents-2026-04-01 beta header required

managed-agents-2026-04-01 beta header required

Permission policies control when agent and MCP tools execute

AGENT TOOLSET

type: agent_toolset_20260401

default_config.permission_policy

default if omitted: always_allow

INDIVIDUAL TOOL OVERRIDE

default_config: always_allow

configs

name: bash

permission_policy: always_ask

PERMISSION POLICIES

always_allow

tool executes automatically

no confirmation

always_ask

session waits for approval

requires user.tool_confirmation

MCP TOOLSET

type: mcp_toolset

mcp_server_name must match mcp_servers name

default: always_ask

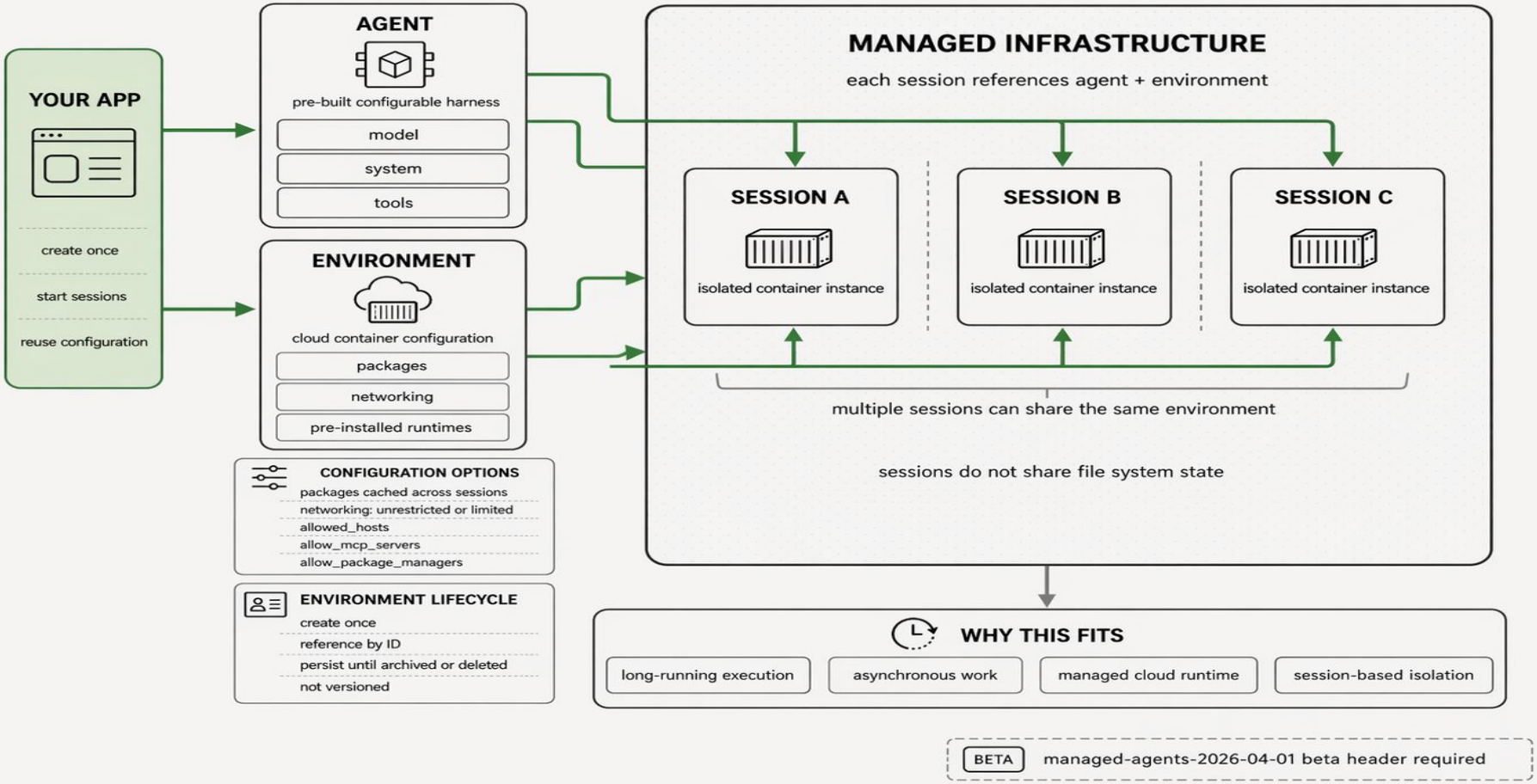
1. **agent.tool_use / agent.mcp_tool_use**
2. **session.status_idle**
   stop_reason: requires_action
3. **user.tool_confirmation**
   result: allow
   result: deny
   deny_message
4. **session returns to running**

CUSTOM TOOLS

not governed by permission policies | agent.custom_tool_use | application decides execution | user.custom_tool_result

Pre-built, configurable agent harness in managed infrastructure

Best for long-running tasks and asynchronous work



Start a session and run agent work inside a live execution context

session = running agent instance within an environment

AGENT

agent ID
latest version by default
can pin a specific version
defines model, system, tools, MCP servers

VAULT AUTH

vault_ids
MCP authentication through vaults
stored OAuth credentials
token refresh managed
session can continue if auth fails
session.error emitted on MCP auth failure

vault_ids

SESSION RESOURCES

A) MEMORY STORE

type: memory_store
memory_store_id
access: read_write or read_only
prompt
persistent memory across sessions
maximum 8 memory stores per session

B) FILES

type: file
file_id
mount_path
upload then mount in container
maximum 100 files per session

C) GITHUB REPOSITORY

type: github_repository
url
mount_path
authorization_token
repository cache speeds future sessions

D) OPTIONAL MCP ACCESS

MCP tools authenticated through vault_ids
remote tool access at session runtime

GITHUB NOTE

clone, read, create pull requests
authorization token not echoed in responses
use minimum required token permissions

ENVIRONMENT

environment ID
created separately
cloud container runtime

SESSION

live execution context
references agent + environment
maintains conversation history
across multiple interactions

EVENT STREAM

USER EVENTS
user.message
user.interrupt
user.custom_tool_result
user.tool_confirmation
user.define_outcome

AGENT EVENTS
agent.message
agent.thinking
agent.tool_use
agent.tool_result
agent.mcp_tool_use
agent.mcp_tool_result
agent.custom_tool_use

SESSION EVENTS
session.status_running
session.status_idle
session.status_rescheduled
session.status_terminated
session.error
session.outcome_evaluated

SPAN EVENTS
span.model_request_start
span.model_request_end
span.outcome_evaluation_start
span.outcome_evaluation_ongoing
span.outcome_evaluation_end

EVENT NOTE

processed_at timestamp
null means queued
open stream before sending events

LIVE CONTROL

send user.message → stream responses → user.interrupt → redirect with new user.message

OUTCOME MODE

user.define_outcome
description
rubric: text or file
max_iterations
grader evaluates each iteration
feedback returns to the agent
session can continue after final evaluation
one outcome at a time

GRADER

provisions automatically
separate context window
per-criterion breakdown
specific gaps feed next iteration

managed-agents-2026-04-01 beta header required
outcomes and memory are research preview features

Container reference

Pre-installed packages, databases, and utilities available in cloud containers

PROGRAMMING LANGUAGES

Language	Version	Package manager
Python	3.12+	pip, uv
Node.js	20+	npm, yarn, pnpm
Go	1.22+	go modules
Rust	1.77+	cargo
Java	21+	maven, gradle
Ruby	3.3+	bundler, gem
PHP	8.3+	composer
C/C++	GCC 13+	make, cmake

CLOUD CONTAINER

pre-installed runtime environment
available immediately
no installation steps

DATABASES

SQLite	Pre-installed, available immediately
PostgreSQL client	psql client for connecting to external databases
Redis client	redis-cli for connecting to external instances

DATABASE NOTE

Database servers are not running in the container by default

The container includes client tools for connecting to external database instances

SQLite is fully available for local use

UTILITIES

SYSTEM TOOLS

- git - Version control
- curl, wget - HTTP clients
- jq - JSON processing
- tar, zip, unzip - Archive tools
- ssh, scp - Remote access (requires network enabled)
- tmux, screen - Terminal multiplexers

DEVELOPMENT TOOLS

- make, cmake - Build systems
- docker - Container management (limited availability)
- ripgrep (rg) - Fast file search
- tree - Directory visualization
- htop - Process monitoring

TEXT PROCESSING

- sed, awk, grep - Stream editors
- vim, nano - Text editors
- diff, patch - File comparison

CONTAINER SPECIFICATIONS

Property	Value
Operating system	Ubuntu 22.04 LTS
Architecture	x86_64 (amd64)
Memory	Up to 8 GB
Disk space	Up to 10 GB
Network	Disabled by default (enable in environment config)

managed-agents-2026-04-01
beta header required